

Survey Communication Analysis

Code Sample for IDA by Katharyn Loweth

2025-02-27

Background: The sample code was pulled from engagement analysis of communication strategies completed as part of data analyst fellowship at the U.S. Census Bureau in June-August 2024. The code specifically analyzes how respondent engagement changes in the time period after planned survey mailings were sent to non-respondents for 5 different establishment-level surveys across 2017-2022 time period.

Disclaimer: Any opinions and conclusions expressed herein are those of the author(s) and do not reflect the views of the U.S. Census Bureau. The Census Bureau has reviewed this data product to ensure appropriate access, use, and disclosure avoidance protection of the confidential CBDRB-FY24-EWD001-007)

The database connection has been changed for external publication. For purposes of code example, setting the eval=FALSE globally so that all code can be viewed in the rendered document.

Preparation

Load packages

```
library(tidyverse)
library(haven)
library(lubridate)
library(ggplot2)
library(DBI)
library(stringr)
```

Setting Up Database Connection

```
pass = rstudioapi::askForPassword("Database password")
con_string = paste0("Driver=driver;DBQ=db;SVD=EXTRACTS;UID=username;PWD=",
  pass)

query_aces <- "select * from ACES"
con = dbConnect(odbc::odbc(), .connection_string = con_string,
  timeout = 10)
aces_query = dbplyr::build_sql(dbplyr::sql(query_aces), con = con)
aces = DBI::dbGetQuery(con, aces_query)
```

Reading in Compendium

File paths have been changed for external release and no longer work.

```
comm = read_csv("inputs/compendium updated/comm_more.csv")
email = read_csv("inputs/compendium updated/email_more.csv")
```

```
head(comm)
head(email)
```

Loading in data

Previous manipulation occurred for the survey data (in exploratory analysis modeling file) that calculated the daily authentication code burn use and ratio for each day in the collection period where code use is greater than 0.

```
# reading in survey data
aces_stats = read.csv("inputs/auth_code_br_duedate/aces_daily_br.csv")
sas_stats = read.csv("inputs/auth_code_br_duedate/sas_daily_br.csv")
awts_stats = read.csv("inputs/auth_code_br_duedate/awts_daily_br.csv")
arts_stats = read.csv("inputs/auth_code_br_duedate/arts_daily_br.csv")
cosasm_stats = read.csv("inputs/auth_code_br_duedate/COSASM_daily_br.csv")

# reading in communication-related data
mail = read.csv("inputs/compendium updated/mail_more.csv")

# reformatting date variable in all datasets
comm$date = as.Date(comm$date, format = "%Y-%m-%d")
mail$date = as.Date(mail$date, format = "%m/%d/%Y")

aces_stats$AUTHDATE_BANK = as.Date(aces_stats$AUTHDATE_BANK,
  format = "%Y-%m-%d")
arts_stats$AUTHDATE_BANK = as.Date(arts_stats$AUTHDATE_BANK,
  format = "%Y-%m-%d")
awts_stats$AUTHDATE_BANK = as.Date(awts_stats$AUTHDATE_BANK,
  format = "%Y-%m-%d")
sas_stats$AUTHDATE_BANK = as.Date(sas_stats$AUTHDATE_BANK, format = "%Y-%m-%d")
cosasm_stats$AUTHDATE_BANK = as.Date(cosasm_stats$AUTHDATE_BANK,
  format = "%Y-%m-%d")

# calculating the number of days mailings were sent for
# each survey collection period splitting the survey id
# into two variables
mail_agg = mail %>%
  group_by(survey_id, date) %>%
  summarise(total_day_mail_sent = sum(n_sent))
mail_agg[c("Survey", "Year")] <- str_split_fixed(mail_agg$survey_id,
  "_", 2)
```

Analysis

Function to look at mail activity window

```
mail_window = function(surveydata, maildata, surveyname, windowlimit) {
  # function reviews the survey data with daily
  # authentication code use and filters the dataset so it
  # only includes the days that are within a certain
  # window of a mailing saves results in new dataframe

  # inputs: surveydata - dataset for survey; unique
```

```

# variable is the AUTHDATE_BANK date because of
# previous manipulation maildata - dataset of mailings
# across all surveys and collection periods surveyname
# - character that matches the 'Survey' field in
# maildata windowlimit - integer that defines the
# window (in days) the function will look after the
# maildate

results_dataset = data.frame()
# filter maildata so it only includes survey years
# 2017-2022 and focuses only on one survey
surveymaildata = maildata %>%
  filter(Survey == surveyname & Year > 2016)
# iterates through the filtered maildata dates
for (i in seq(nrow(surveymaildata))) {
  maildate = surveymaildata$date[i] #date that a mailing was sent out

  # creates a window of time, because maildate is a
  # date datatype adding a number will result in a
  # date in the future
  mailwindow = maildate + windowlimit

  # filters surveydata so that it only includes
  # observations whose date falls within the date
  # window
  filtered_surveydata = surveydata %>%
    filter(AUTHDATE_BANK >= maildate & AUTHDATE_BANK <=
      mailwindow)

  # if statement excludes instances where no codes
  # were used in the period following the mailout
  # date
  if (nrow(filtered_surveydata) > 0) {
    filtered_length = nrow(filtered_surveydata) - 1
    filtered_surveydata$DOW = weekdays(filtered_surveydata$AUTHDATE_BANK)
    # counts the number of days within the window;
    # day 0 is the mailout day
    day_count = data.frame(Day_Count = seq(0, filtered_length))
    # represents the count of the specific mailing
    # in the surveymaildata
    mail_count = data.frame(Mail_Count = rep(i, nrow(filtered_surveydata)))

    # save results filtered surveydataset and the
    # created columns together
    filtered_surveydata_v2 = bind_cols(filtered_surveydata,
      day_count, mail_count)

    # save created dataset for the mailing window
    # in the output dataframe
    results_dataset = bind_rows(results_dataset, filtered_surveydata_v2)
  }
}

```

```

    return(results_dataset)
}

# running function for AWTs and creating graphs
awts_mailperiods = mail_window(awts_stats, mail_agg, "AWTS",
    14)

ggplot(awts_mailperiods %>%
    filter(STATP00 == "2020A1"), aes(x = days_relative_duedate,
    y = count_used, colour = as.factor(Mail_Count))) + geom_line(linewidth = 0.75) +
    theme(legend.position = "none") + geom_vline(xintercept = 0,
    linetype = "dotted", linewidth = 0.75)

ggplot(awts_mailperiods %>%
    filter(STATP00 == "2020A1"), aes(x = days_relative_duedate,
    y = used_overall_ratio, colour = as.factor(Mail_Count))) +
    geom_line(linewidth = 0.75) + theme(legend.position = "none") +
    geom_vline(xintercept = 0, linetype = "dotted", linewidth = 0.75)

# running function for other survey datasets and creating
# similar graphs
arts_mailperiods = mail_window(arts_stats, mail_agg, "ARTS",
    14)
aces_mailperiods = mail_window(aces_stats, mail_agg, "ACES",
    14)
sas_mailperiods = mail_window(sas_stats, mail_agg, "SAS", 14)
cosasm_mailperiods = mail_window(arts_stats, mail_agg, "COS/ASM",
    14)

ggplot(arts_mailperiods %>%
    filter(STATP00 == "2020A1"), aes(x = days_relative_duedate,
    y = count_used, colour = as.factor(Mail_Count))) + geom_line(linewidth = 0.75) +
    theme(legend.position = "none") + geom_vline(xintercept = 0,
    linetype = "dotted", linewidth = 0.75)

ggplot(arts_mailperiods %>%
    filter(STATP00 == "2020A1"), aes(x = days_relative_duedate,
    y = used_overall_ratio, colour = as.factor(Mail_Count))) +
    geom_line(linewidth = 0.75) + theme(legend.position = "none") +
    geom_vline(xintercept = 0, linetype = "dotted", linewidth = 0.75)

ggplot(aces_mailperiods %>%
    filter(STATP00 == "2020A1"), aes(x = days_relative_duedate,
    y = count_used, colour = as.factor(Mail_Count))) + geom_line(linewidth = 0.75) +
    theme(legend.position = "none") + geom_vline(xintercept = 0,
    linetype = "dotted", linewidth = 0.75)

ggplot(aces_mailperiods %>%
    filter(STATP00 == "2020A1"), aes(x = days_relative_duedate,
    y = used_overall_ratio, colour = as.factor(Mail_Count))) +
    geom_line(linewidth = 0.75) + theme(legend.position = "none") +
    geom_vline(xintercept = 0, linetype = "dotted", linewidth = 0.75)

```

```

percent_change = function(dataset) {
  # function calculates the percent change in the
  # authentication use count between days in each mailing
  # window (represented by mail_count) specifically
  # compares value of count_used to the previous
  # day's/observation's value input is the dataset
  # created with the previous function output is dataset
  # with new column 'Percent_change'
  d2 = dataset %>%
    arrange(Mail_Count) %>%
    mutate(Percent_Change = round(((count_used/lag(count_used) -
      1) * 100), 2)) %>%
    select(STATPO0, AUTHDATE_BANK, DOW, count_used, Day_Count,
      Mail_Count, Percent_Change)
  # replacing values for the mailout day since we are not
  # looking outside the mailing window
  d2$Percent_Change[d2$Day_Count == 0] <- NA
  return(d2)
}

# running function with all survey datasets
awts_percchange = percent_change(awts_mailperiods)
arts_percchange = percent_change(arts_mailperiods)
sas_percchange = percent_change(sas_mailperiods)
aces_percchange = percent_change(aces_mailperiods)
cosasm_percchange = percent_change(cosasm_mailperiods)

# creating variable in survey dataset that represent the
# survey it is associated with
cosasm_percchange$Survey = "COS/ASM"
awts_percchange$Survey = "AWTS"
aces_percchange$Survey = "ACES"
arts_percchange$Survey = "ARTS"
sas_percchange$Survey = "SAS"

# compiling results into one dataset
allsurveyemail = rbind(cosasm_percchange, awts_percchange, aces_percchange,
  arts_percchange, sas_percchange)

```

Median Burn Count Visualization

```

# visualization for Median authentication code use across
# all survey mailings
median_mail_count = allsurveyemail %>%
  group_by(Survey, Day_Count) %>%
  summarise(median_usecount = median(count_used, na.rm = TRUE),
    median_percchange = median(Percent_Change, na.rm = TRUE))
cols <- c(ACES = "#26C6DA", ARTS = "#112E51", AWTS = "#FF7043",
  `COS/ASM` = "#78909C", SAS = "#2E78D2")

mail_window = ggplot(data = median_mail_count, aes(x = Day_Count,
  y = median_usecount, color = Survey)) + geom_line(linewidth = 1) +
  scale_color_manual(values = cols) + theme_minimal()

```

```

+ggtitle("Impact of Mailings Occurs 4-7 Days after Sendout")
+theme(text = element_text(size = 14)) + ylab("Median Number of Authentication Codes Burned")
+scale_x_continuous(name = "", limits = c(0, 10), breaks = c(0,
  5, 10), label = c("Mailout Day", "5 Days After", "10 Days After"))
mail_window
ggsave(filename = "Figures/mail_window.png", plot = mail_window,
  width = 8, height = 6, units = "in")

# visualization for median percent change across all survey
# mailings
mail_window_change = ggplot(data = median_mail_count, aes(x = Day_Count,
  y = median_percchange, color = Survey)) + geom_line(linewidth = 0.75) +
  scale_color_manual(values = cols) + theme_minimal()
+ggtitle("Authentication Code Use Percentage Change the 14 days after Mailings",
  subtitle = "For 5 Annual Economic Surveys") + theme(text = element_text(size = 10,
  family = "Roboto")) + ylab("Median Authentication Code Use Count") +
  xlab("Days since Mailing")
mail_window_change

```

Overlap between email and mail

```

# using dataset without the percent change
awts_mailperiods$Survey = "AWTS"
aces_mailperiods$Survey = "ACES"
sas_mailperiods$Survey = "SAS"
cosasm_mailperiods$Survey = "COS/ASM"
arts_mailperiods$Survey = "ARTS"

comm[c("Survey", "Year")] <- str_split_fixed(comm$survey_ID,
  "_", 2)
comm$com_type = toupper(comm$com_type)

email_check = function(surveydata, commdata, surveyname) {
  # function looks through the dates in the surveydata
  # dataset and indicates whether an email was sent on
  # that day based on the survey's communication strategy
  # returns survey dataset with an additional dummy
  # variable

  surveycomm = commdata %>%
    filter(Survey == surveyname & com_type == "EMAIL")
  emaildates = unique(surveycomm$date)
  surveydata$email_ind = ifelse(surveydata$AUTHDATE_BANK %in%
    emaildates, 1, 0)
  return(surveydata)
}

# running mailingperiod surveydata through the function
arts_mailperiods_v2 = email_check(arts_mailperiods, comm, "ARTS")
aces_mailperiods_v2 = email_check(aces_mailperiods, comm, "ACES")
awts_mailperiods_v2 = email_check(awts_mailperiods, comm, "AWTS")
sas_mailperiods_v2 = email_check(sas_mailperiods, comm, "SAS")
cosasm_mailperiods_v2 = email_check(cosasm_mailperiods, comm,
  "COS/ASM")

```

Overlap visualizations

visuals looking at mailing periods and highlights if on that day an email was sent

```
f1 = ggplot(awts_mailperiods_v2 %>%
  filter(STATP00 == "2020A1"), aes(x = days_relative_duedate,
    y = count_used, colour = as.factor(Mail_Count))) + geom_line(linewidth = 0.75) +
  geom_vline(xintercept = 0, linetype = "dotted", linewidth = 0.75)

f1 + geom_point(data = awts_mailperiods_v2 %>%
  filter(STATP00 == "2020A1" & email_ind == 1), aes(x = days_relative_duedate,
    y = count_used, colour = as.factor(Mail_Count)), size = 3) +
  theme_minimal() + theme(legend.position = "none")

f2 = ggplot(arts_mailperiods_v2 %>%
  filter(STATP00 == "2020A1"), aes(x = days_relative_duedate,
    y = count_used, colour = as.factor(Mail_Count))) + geom_line(linewidth = 0.75) +
  geom_vline(xintercept = 0, linetype = "dotted", linewidth = 0.75)

f2 + geom_point(data = arts_mailperiods_v2 %>%
  filter(STATP00 == "2020A1" & email_ind == 1), aes(x = days_relative_duedate,
    y = count_used, colour = as.factor(Mail_Count)), size = 3) +
  theme_minimal() + theme(legend.position = "none")
```

Overlap calculations

looking at the maximum counts in each mailing window and discerning whether that day was also a day an email was sent

```
arts_max = arts_mailperiods_v2 %>%
  group_by(Mail_Count) %>%
  top_n(1, count_used)
table(arts_max$email_ind)
table(arts_max$email_ind, arts_max$STATP00)
```

```
aces_max = aces_mailperiods_v2 %>%
  group_by(Mail_Count) %>%
  top_n(1, count_used)
table(aces_max$email_ind)
table(aces_max$email_ind, aces_max$STATP00)
```

```
awts_max = awts_mailperiods_v2 %>%
  group_by(Mail_Count) %>%
  top_n(1, count_used)
table(awts_max$email_ind)
table(awts_max$email_ind, awts_max$STATP00)
```

```
sas_max = sas_mailperiods_v2 %>%
  group_by(Mail_Count) %>%
  top_n(1, count_used)
table(sas_max$email_ind)
table(sas_max$email_ind, sas_max$STATP00)
```

```
cosasm_max = cosasm_mailperiods_v2 %>%
  group_by(Mail_Count) %>%
  top_n(1, count_used)
```

```
table(cosasm_max$email_ind)
table(cosasm_max$email_ind, cosasm_max$STATP00)
```

calculating the percentage of the mailing windows (14 days) for each survey collection period where the day in the window with the highest count was also the day where the email was sent

```
aces_max_2 = aces_max %>%
  group_by(Survey, STATP00) %>%
  summarise(yearcount = n(), email_max_count = sum(email_ind),
    email_max_perc = (round(email_max_count/yearcount * 100,
      2)))
awts_max_2 = awts_max %>%
  group_by(Survey, STATP00) %>%
  summarise(yearcount = n(), email_max_count = sum(email_ind),
    email_max_perc = (round(email_max_count/yearcount * 100,
      2)))
arts_max_2 = arts_max %>%
  group_by(Survey, STATP00) %>%
  summarise(yearcount = n(), email_max_count = sum(email_ind),
    email_max_perc = (round(email_max_count/yearcount * 100,
      2)))
sas_max_2 = sas_max %>%
  group_by(Survey, STATP00) %>%
  summarise(yearcount = n(), email_max_count = sum(email_ind),
    email_max_perc = (round(email_max_count/yearcount * 100,
      2)))
cosasm_max_2 = cosasm_max %>%
  group_by(Survey, STATP00) %>%
  summarise(yearcount = n(), email_max_count = sum(email_ind),
    email_max_perc = (round(email_max_count/yearcount * 100,
      2)))

allmax = rbind(aces_max_2, awts_max_2, arts_max_2, sas_max_2,
  cosasm_max_2)

allmax$Year[allmax$STATP00 == "2017A1"] <- 2017
allmax$Year[allmax$STATP00 == "2018A1"] <- 2018
allmax$Year[allmax$STATP00 == "2019A1"] <- 2019
allmax$Year[allmax$STATP00 == "2020A1"] <- 2020
allmax$Year[allmax$STATP00 == "2021A1"] <- 2021
allmax$Year[allmax$STATP00 == "2022A1"] <- 2022

# calculating mean for different years
mean(allmax$email_max_perc[allmax$Year == 2022])
mean(allmax$email_max_perc[allmax$Year == 2018])

# looking at a smaller window (days 3 to 7 after mailout)
allmailperiods = rbind(cosasm_mailperiods_v2, aces_mailperiods_v2,
  awts_mailperiods_v2, arts_mailperiods_v2, sas_mailperiods_v2)
allmail_days3to7 = allmailperiods %>%
  filter(Day_Count <= 7 & Day_Count > 2)
allmail_oneweekmax = allmail_days3to7 %>%
  group_by(Survey, Mail_Count) %>%
  top_n(1, count_used)
```



```

allmail_oneweek_emailmax = allmail_oneweek %>%
  group_by(Survey, STATP00) %>%
  summarise(yearcount = n(), email_max_count = sum(email_ind),
    email_max_perc = (round(email_max_count/yearcount * 100,
      2)))

allmail_oneweek_emailmax$Year[allmail_oneweek_emailmax$STATP00 ==
  "2017A1"] <- 2017
allmail_oneweek_emailmax$Year[allmail_oneweek_emailmax$STATP00 ==
  "2018A1"] <- 2018
allmail_oneweek_emailmax$Year[allmail_oneweek_emailmax$STATP00 ==
  "2019A1"] <- 2019
allmail_oneweek_emailmax$Year[allmail_oneweek_emailmax$STATP00 ==
  "2020A1"] <- 2020
allmail_oneweek_emailmax$Year[allmail_oneweek_emailmax$STATP00 ==
  "2021A1"] <- 2021
allmail_oneweek_emailmax$Year[allmail_oneweek_emailmax$STATP00 ==
  "2022A1"] <- 2022

mean(allmail_oneweek_emailmax$email_max_perc[allmail_oneweek_emailmax$Year ==
  2022])
mean(allmail_oneweek_emailmax$email_max_perc[allmail_oneweek_emailmax$Year ==
  2018])

# looking at the percentage of 14-day mailing windows where
# at least 1 email occurs in the mailing period
email2weeks = allmailperiods %>%
  group_by(Survey, Mail_Count) %>%
  summarise(email_count = sum(email_ind))

prop.table(table(email2weeks$email_count))

# looking at the percentage of 3-7 day mailing windows
# where at least 1 email occurs in the mailing period
email3to7days = allmail_days3to7 %>%
  group_by(Survey, Mail_Count) %>%
  summarise(email_count = sum(email_ind))
prop.table(table(email3to7days$email_count))

# subsetting data to only include mailing windows where
# emails were sent during days 3-7 after mailout
allmail_days3to7$survey_mail = paste(allmail_days3to7$Survey,
  allmail_days3to7$Mail_Count, sep = "_")
email3to7days$survey_mail = paste(email3to7days$Survey, email3to7days$Mail_Count,
  sep = "_")
emailincluded = unique(email3to7days$survey_mail[email3to7days$email_count >
  0])

# of these only email cases, calculating the percentage
# where the highest use count day was an email day
onlyemailcases = allmail_days3to7 %>%
  filter(survey_mail %in% emailincluded)
onlyemailcases_max = onlyemailcases %>%
  group_by(Survey, Mail_Count) %>%

```

```

    top_n(1, count_used)

table(onlyemailcases_max$email_ind)
prop.table(table(onlyemailcases_max$email_ind))

# subsetting data to only include mailing windows where
# emails were sent within 2 weeks after mailout
allmailperiods$survey_mail = paste(allmailperiods$Survey, allmailperiods$Mail_Count,
    sep = "_")
email2weeks$survey_mail = paste(email2weeks$Survey, email2weeks$Mail_Count,
    sep = "_")
emailincluded_2wk = unique(email2weeks$survey_mail[email2weeks$email_count >
    0])

# of these only email cases, calculating the percentage
# where the highest use count day was an email day
onlyemailcases_2wk = allmailperiods %>%
    filter(survey_mail %in% emailincluded_2wk)
onlyemailcases_2wkmax = onlyemailcases_2wk %>%
    group_by(Survey, Mail_Count) %>%
    top_n(1, count_used)
table(onlyemailcases_2wkmax$email_ind)
prop.table(table(onlyemailcases_2wkmax$email_ind))

```